WAYGROUND formerly Quizizz **Worksheets**

P4B - R Chapter 1 Quiz

Total questions: 10

Worksheet time: 10mins

Name

Class

Date

1. Running code in R language can be done in both **R** and **RStudio**; however, **R** is more comprehensive, as it provides built-in tools for debugging, creating visualisations and offers a more user-friendly interface.

   a) TRUE
   b) FALSE

2. A student executed the following line of code:
   example(rowMeans)
   What does this code do?

   a) Computes the mean of each row in a dataset using the RowMeans function
   b) Displays the help documentation for the RowMeans function
   c) Runs and displays example code demonstrating how to use the RowMeans function
   d) None of the options

3. A student created the following vector in R:
   grades_last_semester <- c(16, 15, 13, 17, 20, 14)
   Which of the following lines of code correctly assigns labels to each element of the vector?

   a) labels(grades_last_semester) <- c("Math Analysis I", "Information Systems", "Business Management", "Economic Analysis", "Computation I", "Linear Algebra")
   b) names(grades_last_semester) <- c("Math Analysis I", "Information Systems", "Business Management", "Economic Analysis", "Computation I")
   c) colnames(grades_last_semester) <- c("Math Analysis I", "Information Systems", "Business Management", "Economic Analysis", "Computation I", "Linear Algebra")
   d) names(grades_last_semester) <- c("Math Analysis I", "Information Systems", "Business Management", "Economic Analysis", "Computation I", "Linear Algebra")

4. Consider the following R code: **grades <- matrix(c(grades_last_semester, grades_this_semester, 6)**
   Assume that both grades_last_semester and grades_this_semester are numeric vectors containing 6 values each. How will the values from these vectors be stored inside the matrix grades?

   a) The values are filled column by column, with grades_last_semester forming the first column and grades_this_semester the second column
   b) The values are filled row by row, with grades_last_semester forming the first row and grades_this_semester the second row
   c) The two vectors are combined into a single row with 12 columns (a 1x12 matrix)
   d) This code gives an error because we are not specifying the desired number of columns.

5. Consider the following matrix: **scores <- matrix(1:12, nrow = 3)**
   What is the result of the following code?
   scores[c(1, 3), 2:3]

   a) [,1] [,2]
      [1,] 2 3
      [2,] 8 9
   b) [,1] [,2]
      [1,] 4 7
      [2,] 6 9
   c) [,1] [,2]
      [1,] 5 8
      [2,] 6 11
   d) [,1] [,2]
      [1,] 4 7
      [2,] 10 11

6. In R, a matrix can only contain elements of the same data type, while a data frame can contain different data types in different columns.

   a) TRUE
   b) FALSE

7. A student creates the following data frame in R:
   students <- data.frame(
   name = c("Ana", "Ben", "Chris"),
   math = c(15, 18, 14),
   science = c(17, 16, 19))
   What is the result of the following command?
   students[students$math > 15, c("name", "science")]

   a) name science
      Ben 16
      Chris 19
   b) name science
      Ben 16
   c) name math
      Ben 18
   d) name science
      1 Ana 17
      2 Ben 16

8. A student runs the following code in R: colours <- factor(c("red", "blue", "green", "red", "blue"))
   What will the output of levels(colours) be?

   a) "blue" "green" "red" "blue" "red"
   b) "blue" "blue" "green" "red" "red"
   c) "red" "blue" "green"
   d) "red" "blue" "green" "red" "blue"

9. In R, a list can contain elements of different types, including numbers, character strings, vectors, and even other lists.

   a) TRUE
   b) FALSE

10. A student runs the following R code:
    my_list <- list(
    matrix1 = matrix(1:4, nrow = 2),
    vector1 = c(10, 20, 30),
    text = "hello"
)
    my_list$matrix1 <- NULL
    Which statement is true about **my_list** after executing this code?

    a) The element matrix1 is completely removed from the list
    b) The element matrix1 becomes a 0×0 matrix
    c) The element matrix1 is replaced with NA
    d) The element matrix1 becomes a list containing NULL

Answer Keys

1. b) FALSE
2. c) Runs and displays example code demonstrating how to use the RowMeans function
3. d) names(grades_last_semester) <- c("Math Analysis I", "Information Systems", "Business Management", "Economic Analysis", "Computation I", "Linear Algebra")
4. a) The values are filled column by column, with grades_last_semester forming the first column and grades_this_semester the second column
5. b) [,1] [,2] [1,] 4 7 [2,] 6 9
6. a) TRUE
7. b) name science Ben 16
8. c) "red" "blue" "green"
9. a) TRUE
10. a) The element matrix1 is completely removed from the list